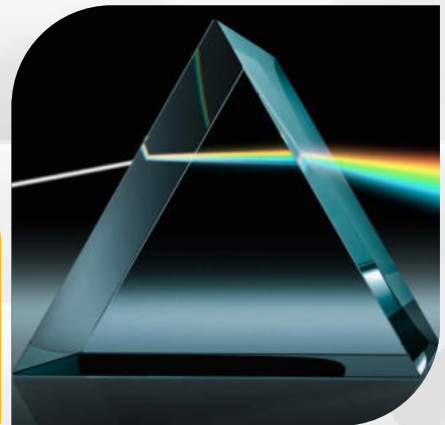


# UNIT 10

## GENERAL WAVES PROPERTIES



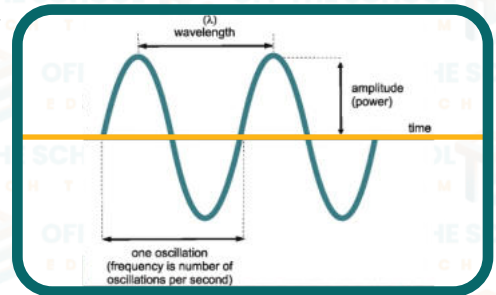
## WAVES

Waves are a fundamental concept in physics that describe the transfer of energy and information through a medium or space

### FORMATION OF WAVES

Waves are formed when a disturbance or energy input is applied to a medium. This disturbance can be caused by various factors, such as the motion of an object, changes in pressure, or vibrations. Once the initial disturbance occurs, it creates a ripple effect, where neighboring particles of the medium interact with each other, transmitting the disturbance outward in all directions.

These oscillations can be either transverse or longitudinal, depending on the type of wave.



### WATER WAVE IN A RIPPLE TANK

A water wave in a ripple tank is a visible pattern formed on the water's surface when it is disturbed. The tank is a shallow container with water, and an overhead light source shines light onto the water. The resulting light and dark regions on a screen or tank's bottom allow observation and study of wave properties like wavelength, frequency, amplitude, and wave speed. The ripple tank is a valuable tool for understanding wave behavior in a controlled setting.

### TRANSVERSE WAVES

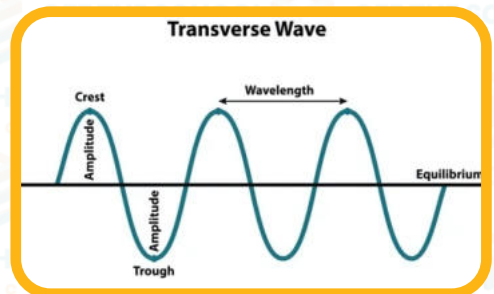
Transverse waves are a type of wave in which the particles of the medium move perpendicular (at right angles) to the direction in which the wave is propagating.

#### REGIONS OF TRANSVERSE WAVES

**1. Amplitude:** The amplitude of a transverse wave is the maximum displacement or distance that a particle in the medium moves from its equilibrium position.

**2. Crest:** The crest is the highest point or peak of a transverse wave.

**3. Trough:** It is the point on the wave where the displacement of the particles is maximum in the downward direction from their equilibrium position.



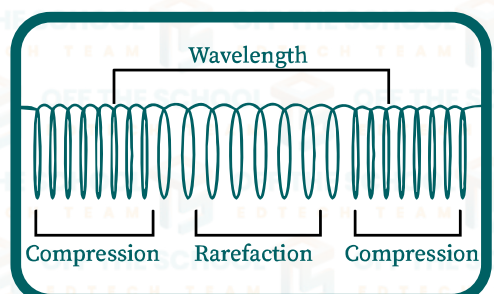
### LONGITUDINAL WAVES

Longitudinal waves are a type of wave in which the particles of the medium move parallel to the direction in which the wave is propagating.

#### REGIONS OF LONGITUDINAL WAVES

**1. Compression:** A compression is a region in a longitudinal wave where the particles of the medium are close together, resulting in a high-pressure area. It corresponds to the regions of maximum particle density.

**2. Rarefaction:** A rarefaction is a region in a longitudinal wave where the particles of the medium are spread out, resulting in a low-pressure area. It corresponds to the regions of minimum particle density.





## DIFFERENCE BETWEEN ELECTROMAGNETIC WAVES AND MECHANICAL WAVE

|                           | Electromagnetic Waves           | Mechanical Waves                   |
|---------------------------|---------------------------------|------------------------------------|
| <b>Nature of waves</b>    | Transverse Oscillating fields   | Longitudinal/ Transverse Waves     |
| <b>Medium requirement</b> | No medium needed                | Require a Medium                   |
| <b>Propagation</b>        | Can travel through vacuum       | Need a physical medium             |
| <b>Speed</b>              | Speed of light in vacuum        | Speed depends on medium            |
| <b>Interaction</b>        | Interact with charged particles | Interact with medium               |
| <b>Polarization</b>       | Can be polarized                | Polarization possible (transverse) |
| <b>Examples</b>           | Radio waves, light, X-rays      | Sound, water, seismic water        |

## PROPERTIES OF WAVES

Properties of waves are specific characteristics that describe the behavior and nature of a wave. These properties help us understand how waves propagate, interact, and transfer energy through a medium or space. Some important properties of waves include:

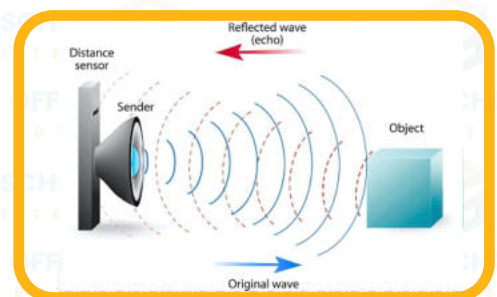
**Reflection:** Reflection is the bouncing back of a wave when it encounters a boundary or obstacle.

**Refraction:** Refraction is the bending of a wave as it passes from one medium to another with different propagation speeds.

**Diffraction:** Diffraction is the bending and spreading out of a wave as it encounters an obstacle or passes through an opening (aperture) in a barrier.

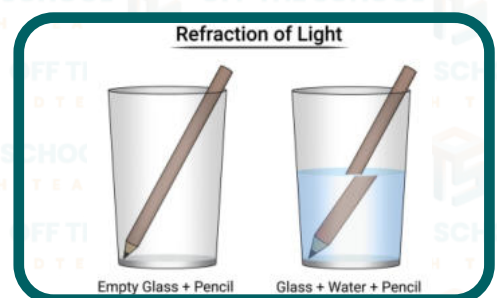
## REFLECTION OF WAVES

Reflection is the phenomenon in which a wave encounters a boundary or obstacle and bounces back, returning to the same medium from which it originated. When a wave reflects, it follows the law of reflection, which states that the angle of incidence (the angle between the incident wave and the normal to the boundary) is equal to the angle of reflection (the angle between the reflected wave and the normal).



## REFRACTION OF WAVES

Refraction is the bending of a wave as it passes from one medium to another with different propagation speeds. When a wave enters a medium with a different speed, its direction changes, and it continues to propagate through the new medium. The degree of bending depends on the change in wave speed between the two media and the angle of incidence relative to the boundary between them.



**DIFFRACTION OF WAVES**

Diffraction is the bending and spreading out of a wave as it encounters an obstacle or passes through an opening (aperture) in a barrier. The extent of diffraction depends on the wavelength of the wave and the size of the obstacle or aperture. When the obstacle's size is comparable to or smaller than the wavelength, significant diffraction occurs. Diffraction is a characteristic property of waves and is observed in various contexts, such as sound waves bending around corners.

**CHARACTERISTICS OF WAVE**

These characteristics help us understand how waves propagate, interact, and transfer energy through a medium or space. Some important properties of waves include:

**Wavelength ( $\lambda$ ):** The wavelength is the distance between two successive points in a wave that are in phase. For transverse waves, it is the distance between two consecutive crests or troughs. For longitudinal waves, it is the distance between two consecutive compressions or rarefactions.

**Frequency ( $f$ ):** The frequency of a wave is the number of complete oscillations or cycles that occur in one second. It is measured in hertz (Hz).

**Time Period ( $t$ ):** The period of a wave is the time it takes to complete one full oscillation or cycle. It is the reciprocal of the frequency and is measured in seconds (s).

**Wave Speed ( $v$ ):** The wave speed is the rate at which the wave propagates through the medium.

**Wave Front:** A wave front is an imaginary surface that connects all points of a wave that are in the same phase.

**PERIODIC MOTION**

A motion repeating itself in an equal time interval is referred to as periodic or oscillatory motion.

**SIMPLE HARMONIC MOTION (SHM)**

When an object oscillates about a fixed position (mean position) its acceleration is directly proportional to its displacement from the mean position and is always directed towards the mean position, its motion is called SHM.

$$a \propto -x$$

$$a = -kx$$

Where  $k$  is spring constant.

Many phenomena include electromagnetic waves, alternating current circuits, musical instruments, bridges, and molecular motion that executes the simple harmonic motion.

**SIMPLE PENDULUM**

A simple pendulum is a mass attached to a string or rod that swings back and forth under the influence of gravity.

**FORCE ON A DISPLACED PENDULUM**

When the bob of the pendulum is displaced at a small angle ( $\theta$ ) to an extreme position.

Two forces that act upon a displaced pendulum:

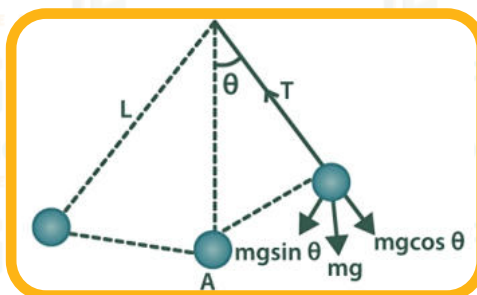
1. Tension 'T' force along the string or rod.
2. Weight  $W = mg$ , acting vertically downward.

The weight is further resolved into its component  $mg \sin \theta$  and  $mg \cos \theta$ .



## PROOF OF MOTION OF SIMPLE PENDULUM IS SHM

A simple pendulum consists of a point mass suspended by a weightless string from a rigid support.



When the bob is displaced by a distance "x" to position "A," work is done ( $Fd = Fx$ ) to restore the bob with a force equal in magnitude but opposite in direction to the applied force "F."

At position "A," the bob gains maximum potential energy before being released, initiating periodic motion around the mean position "O." The potential energy is greatest at points "A" and "B," while the kinetic energy is maximum at position "O" due to the highest velocity.

As the bob moves from "O" to "A" or "O" to "B," its velocity decreases, resulting in a negative acceleration, which becomes zero at the extreme position. Thus, the acceleration of the bob is directly proportional to the displacement but in the opposite direction. Consequently, the motion of a simple pendulum follows Simple Harmonic Motion (SHM).

## TIME PERIOD OF SIMPLE PENDULUM

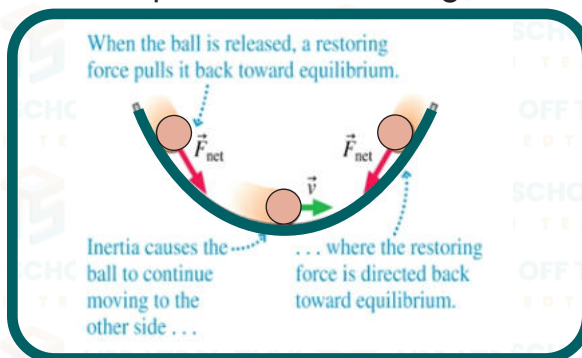
The time period of simple pendulum can be calculate as follow:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Where L is the length of the simple pendulum & g= acceleration due to gravity

## THE BALL AND BOWL SYSTEM AND SHM

In a ball and bowl system, a ball placed in a bowl-shaped hollow move back and forth inside when displaced. This motion is Simple Harmonic Motion (SHM), like a swinging pendulum. The ball experiences a force pulling it back to the center, proportional to its displacement, creating repetitive back-and-forth motion. Many natural systems exhibit SHM, making it an important concept in understanding various phenomena.



## DAMPED SYSTEM

A damped system is a mechanical system that experiences a dissipation of energy, causing the amplitude of its oscillations to decrease over time.

## DAMPED OSCILLATION

A damped system is a mechanical system that experiences a dissipation of energy, causing the amplitude of its oscillations to decrease over time.